

8(a)	any four from: <i>outer / inner membranes</i> 1 allow entry of, oxygen for oxidative phosphorylation / pyruvate for link reaction ; I Krebs cycle 2 allow exit of carbon dioxide from, Krebs cycle / link reaction ; 3 compartmentalisation / described ; <i>inner membrane</i> 4 (location of) ETC to release energy to pump protons into intermembrane space ; 5 (location of) ATP synthase for production of ATP ; 6 cristae / infoldings, to increase surface area for many, ETCs / ATP synthases ; 7 impermeable to, hydrogen ions / protons, to maintain proton gradient or impermeable to, hydrogen ions / protons, to maintain, high H⁺ concentration in intermembrane space ; 8 site of, oxidative phosphorylation / chemiosmosis ;	4
8(b)(i)	any two from: 1 <i>ref. to</i> cells, metabolic rate / energy needs ; 2 in cells about to undergo, mitosis / cell division ; 3 AVP ; e.g. to replace damaged mitochondria	2
8(b)(ii)	hydrolysed / bond broken, (to release energy) or G proteins bind to GTP rather than to ATP ;	1
8(c)	1 degraded / broken down by, lysosomes / hydrolytic enzymes ; 2 components of breakdown / AW, used to produce new mitochondria ;	2